LAPORAN PRAKTIKUM INTERNET OF THINGS

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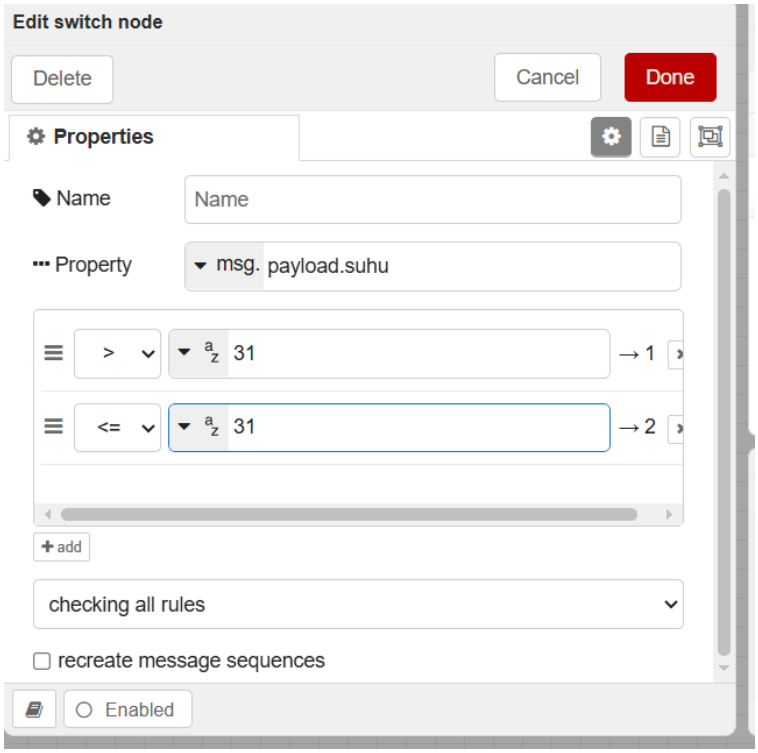
POLITEKNIK NEGERI LAMPUNG

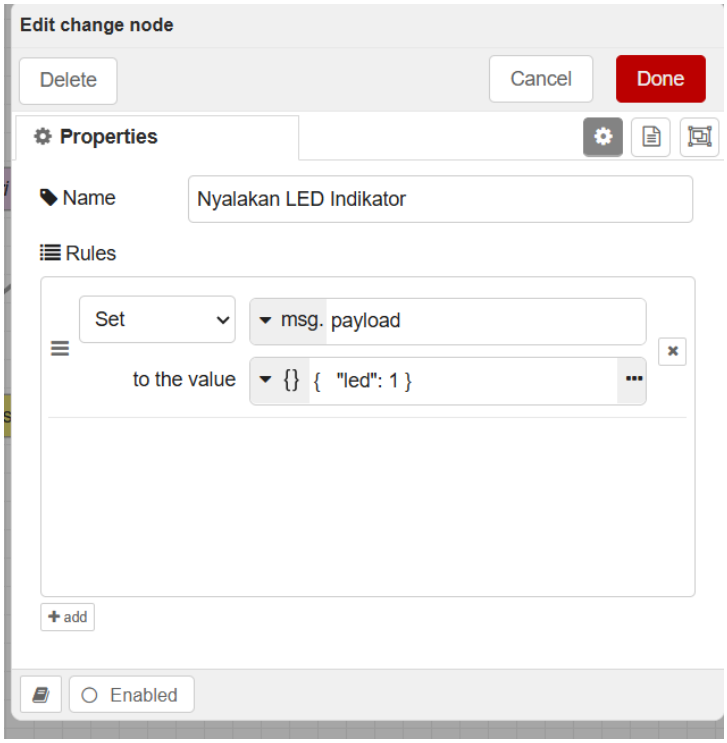
JURUSAN TEKNOLOGI INFORMASI

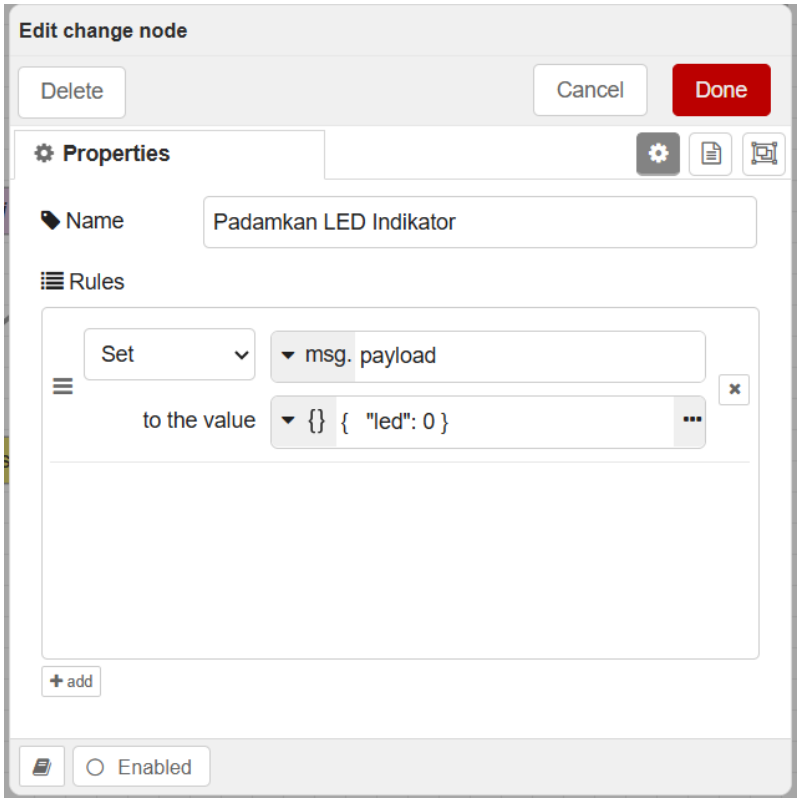
PROGRAM STUDI TEKNOLOGI REKAYASA INTERNET

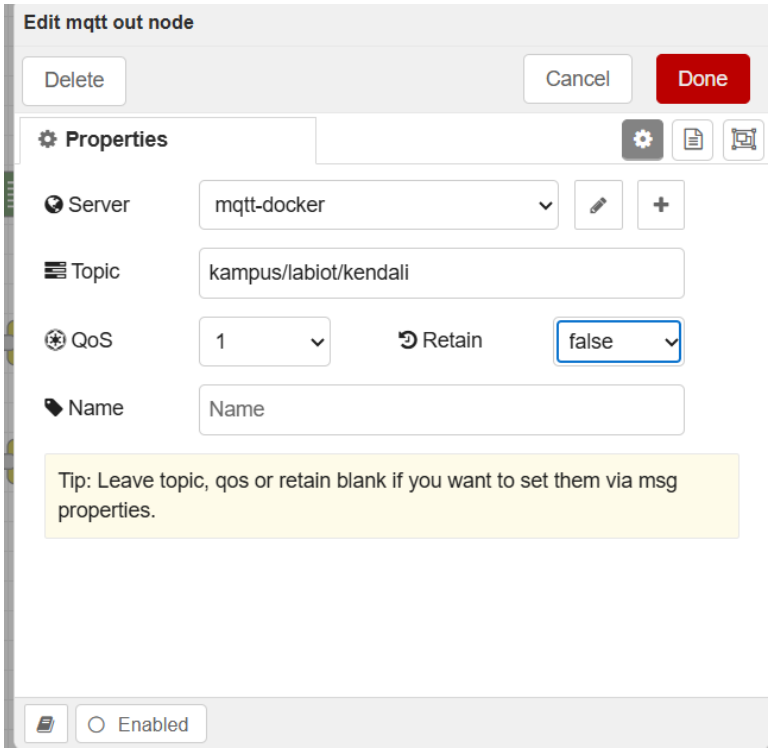
2026

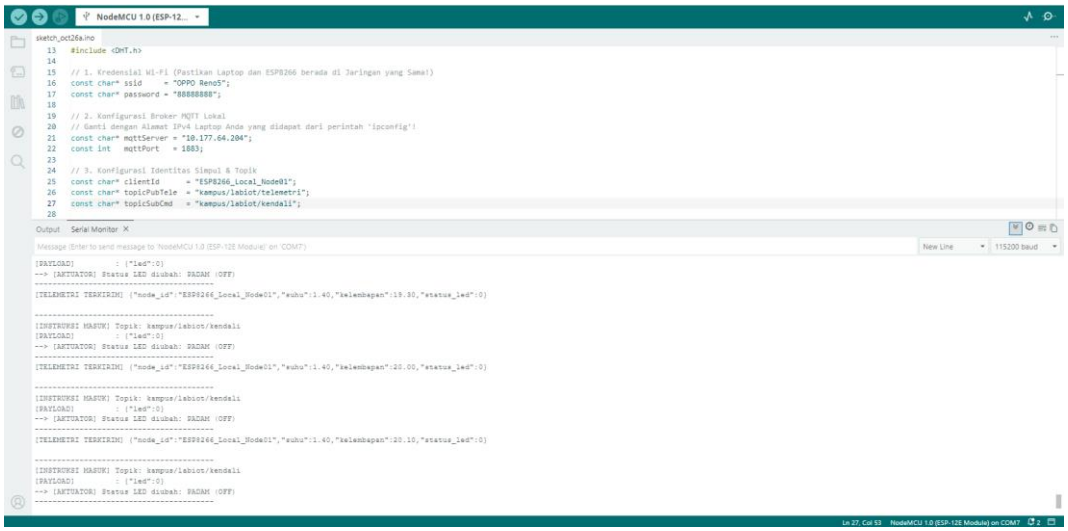
7.5.4 Praktikum 4: Implementasi Rule Engine Otomasi Aktuator Loop Tertutup

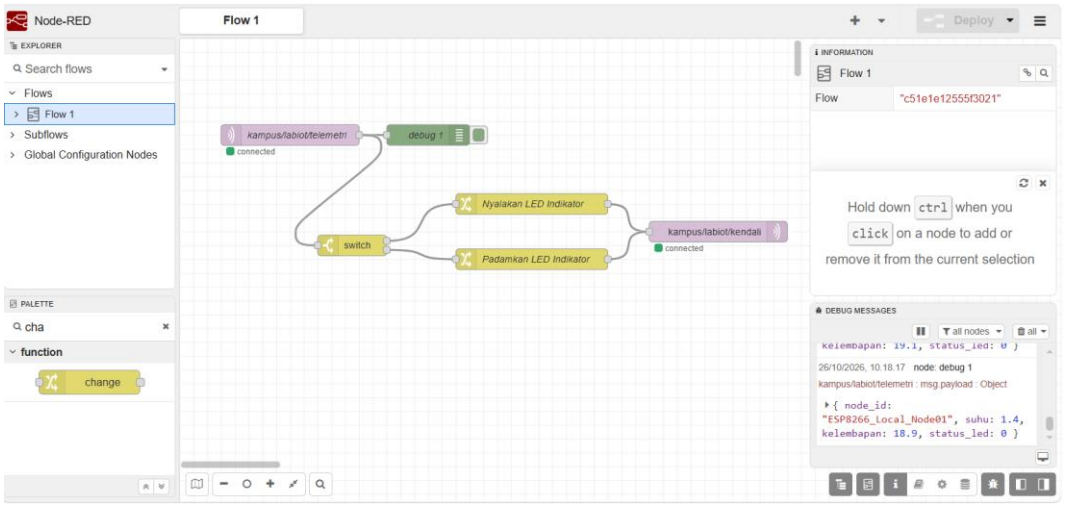





Edit change node
Delete
Cancel
Done
Properties
Name
Padamkan LED Indikator
Rules
Set
msg. payload
to the value
{} { "led": 0 }
add
Enabled


Edit mqtt out node
Delete
Cancel
Done
Properties
Server
mqtt-docker
Topic
kampus/labiot/kendali
QoS
1
Retain
false
Name
Name
Tip: Leave topic, qos or retain blank if you want to set them via msg properties.
Enabled


NodeMCU 1.0 (ESP-12...
sketch_oct26a.ino
#include <DHT.h>
// 1. Kredensial Wi-Fi (Pastikan Laptop dan ESP8266 berada di Jaringan yang Sama!)
const char* ssid = "OPPO Reno5";
const char* password = "88888888";
// 2. Konfigurasi Broker MQTT Lokal
// Ganti dengan Alamat IPv4 Laptop Anda yang didapat dari perintah 'ipconfig'!
const char* mqttServer = "10.177.64.204";
const int mqttPort = 1883;
// 3. Konfigurasi Identitas Simpul & Topik
const char* clientId = "ESP8266_Local_Node01";
const char* topicPubTele = "kampus/labiot/telemetri";
const char* topicSubCmd = "kampus/labiot/kendali";
Output
Serial Monitor
New Line
115200 baud
[PAYLOAD] : {"led":0}
--> [AKTUATOR] Status LED diubah: PADAM (OFF)
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":1.40,"kelembapan":19.80,"status_led":0}
[INSTRUKSI MASUK] Topik: kampus/labiot/kendali
[PAYLOAD] : {"led":0}
--> [AKTUATOR] Status LED diubah: PADAM (OFF)
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":1.40,"kelembapan":20.00,"status_led":0}
[INSTRUKSI MASUK] Topik: kampus/labiot/kendali
[PAYLOAD] : {"led":0}
--> [AKTUATOR] Status LED diubah: PADAM (OFF)
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":1.40,"kelembapan":20.10,"status_led":0}
[INSTRUKSI MASUK] Topik: kampus/labiot/kendali
[PAYLOAD] : {"led":0}
--> [AKTUATOR] Status LED diubah: PADAM (OFF)
Ln 27, Col 53 NodeMCU 1.0 (ESP-12E Module) on COM7


Node-RED
Flow 1
Deploy
EXPLORER
Search flows
Flows
Flow 1
Subflows
Global Configuration Nodes
kampus/labiot/telemetri
connected
debug 1
switch
Nyalakan LED Indikator
Padamkan LED Indikator
kampus/labiot/kendali
connected
INFORMATION
Flow 1
Flow
"c51e1e12555f3021"
Hold down ctrl when you click on a node to add or remove it from the current selection
DEBUG MESSAGES
all nodes
all
kelembapan: 19.1, status_led: 0 }
26/10/2026, 10.18.17 node: debug 1
kampus/labiot/telemetri : msg.payload : Object
{ node_id: "ESP8266_Local_Node01", suhu: 1.4, kelembapan: 18.9, status_led: 0 }
PALETTE
cha
function
change